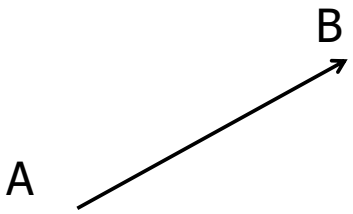
Lecture 4
Vector Spaces

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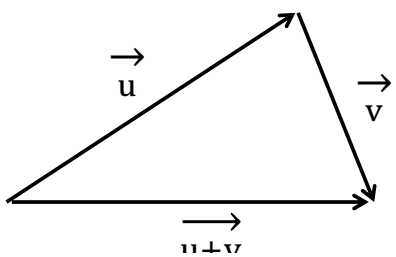
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4.0 Vectors as we have learned in highschool

- An oriented interval of a straight line:



- Used to represent *speed of motions, forces,....*
- Addition:



4.1 Vectors in R^n

- An ordered *n*-tuple:

  a sequence of *n* real number $(x_1, x_2, \cdots, x_n)$

- *n*-space: R^n

  the set of all ordered n-tuple

We call a sequence of *n* real number $(x_1, x_2, \cdots, x_n)$ a ***vector*** of R^n

- Ex:

$n = 1$ R^1 = 1-space
= set of all real number

$n = 2$ R^2 = 2-space
= set of all ordered pair of real numbers (x_1, x_2)

$n = 3$ R^3 = 3-space
= set of all ordered triple of real numbers (x_1, x_2, x_3)

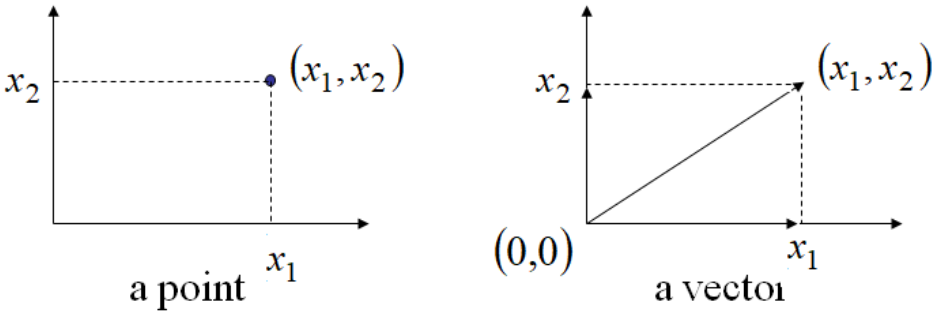
$n = 4$ R^4 = 4-space
= set of all ordered quadruple of real numbers (x_1, x_2, x_3, x_4)

- Notes:

(1) An n-tuple $(x_1, x_2, \cdots, x_n)$ can be viewed as a point in R^n with the x_i's as its coordinates.

(2) An n-tuple $(x_1, x_2, \cdots, x_n)$ can be viewed as a vector $x = (x_1, x_2, \cdots, x_n)$ in R^n with the x_i's as its components.

- Ex:



$\mathbf{u} = (u_1, u_2, \cdots, u_n)$, $\mathbf{v} = (v_1, v_2, \cdots, v_n)$ (two vectors in R^n)

- Equal:

$\mathbf{u} = \mathbf{v}$ if and only if $u_1 = v_1,\ u_2 = v_2, \cdots, u_n = v_n$

- Vector addition (the sum of **u** and **v**):

$\mathbf{u} + \mathbf{v} = (u_1 + v_1, u_2 + v_2, \cdots, u_n + v_n)$

- Scalar multiplication (the scalar multiple of **u** by c):

$c\mathbf{u} = (cu_1, cu_2, \cdots, cu_n)$

- Notes:

The sum of two vectors and the scalar multiple of a vector in R^n are called *the standard operations* in R^n.

- Negative:

$$-\mathbf{u} = (-u_1, -u_2, -u_3, \ldots, -u_n)$$

- Difference:

$$\mathbf{u} - \mathbf{v} = (u_1 - v_1, u_2 - v_2, u_3 - v_3, \ldots, u_n - v_n)$$

- Zero vector:

$$\mathbf{0} = (0, 0, \ldots, 0)$$

- Notes:

(1) The zero vector $\mathbf{0}$ in R^n is called the **additive identity** in R^n.

(2) The vector $-\mathbf{v}$ is called the **additive inverse** of $\mathbf{v}$.

- Thm 4.1: (Properties of vector addition and scalar multiplication)

Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be vectors in R^n, and let c and d be scalars.

(1) $\mathbf{u}+\mathbf{v}$ is a vector in R^n

(2) $\mathbf{u}+\mathbf{v} = \mathbf{v}+\mathbf{u}$

(3) $(\mathbf{u}+\mathbf{v})+\mathbf{w} = \mathbf{u}+(\mathbf{v}+\mathbf{w})$

(4) $\mathbf{u}+\mathbf{0} = \mathbf{u}$

(5) $\mathbf{u}+(-\mathbf{u}) = \mathbf{0}$

(6) $c\mathbf{u}$ is a vector in R^n

(7) $c(\mathbf{u}+\mathbf{v}) = c\mathbf{u}+c\mathbf{v}$

(8) $(c+d)\mathbf{u} = c\mathbf{u}+d\mathbf{u}$

(9) $c(d\mathbf{u}) = (cd)\mathbf{u}$

(10) $1(\mathbf{u}) = \mathbf{u}$

- Ex 5: (Vector operations in R^4)

Let $\mathbf{u}=(2,-1,5,0)$, $\mathbf{v}=(4,3,1,-1)$, and $\mathbf{w}=(-6,2,0,3)$ be vectors in R^4.
Find $\mathbf{x}$ in each of the following equations.

(a) $\mathbf{x}=2\mathbf{u}-(\mathbf{v}+3\mathbf{w})$

(b) $3(\mathbf{x}+\mathbf{w})=2\mathbf{u}-\mathbf{v}+\mathbf{x}$

Sol: (a) $\mathbf{x}=2\mathbf{u}-(\mathbf{v}+3\mathbf{w})$

$=2\mathbf{u}-\mathbf{v}-3\mathbf{w}$

$=(4,-2,10,0)-(4,3,1,-1)-(-18,6,0,9)$

$=(4-4+18,-2-3-6,10-1-0,0+1-9)$

$=(18,-11,9,-8).$

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(b) $3(\mathbf{x}+\mathbf{w})=2\mathbf{u}-\mathbf{v}+\mathbf{x}$

$3\mathbf{x}+3\mathbf{w}=2\mathbf{u}-\mathbf{v}+\mathbf{x}$

$3\mathbf{x}-\mathbf{x}=2\mathbf{u}-\mathbf{v}-3\mathbf{w}$

$2\mathbf{x}=2\mathbf{u}-\mathbf{v}-3\mathbf{w}$

$\mathbf{x}=\mathbf{u}-\frac{1}{2}\mathbf{v}-\frac{3}{2}\mathbf{w}$

$=(2,1,5,0)+\left(-2,\frac{-3}{2},\frac{-1}{2},\frac{1}{2}\right)+\left(9,-3,0,\frac{-9}{2}\right)$

$=\left(9,\frac{-11}{2},\frac{9}{2},-4\right)$

- Thm 4.2: (Properties of additive identity and additive inverse)

Let **v** be a vector in R^n and c be a scalar. Then the following is true.

(1) The additive identity is unique. That is, if **u**+**v**=**v**, then **u** = **0**

(2) The additive inverse of **v** is unique. That is, if **v**+**u**=**0**, then **u** = –**v**

(3) 0**v**=**0**

(4) c**0**=**0**

(5) If c**v**=**0**, then c=0 or **v**=**0**

(6) –(– **v**) = **v**

- Linear combination:

The vector **x** is called a ***linear combination*** of $\mathbf{v}_1, \mathbf{v}_2, ..., \mathbf{v}_n$ if it can be expressed in the form

$$\mathbf{x} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n \qquad c_1, c_2, \cdots, c_n : \text{scalar}$$

- Ex 6:

Given **x** = (– 1, – 2, – 2), **u** = (0,1,4), **v** = (– 1,1,2), and **w** = (3,1,2) in R^3. Show that **x** is a linear combination **of u, v** and **w**. That is, we need to find a, b, and c such that **x** = a**u**+b**v**+c**w**.

Sol:

$$\begin{aligned} -b + 3c &= -1 \\ a + b + c &= -2 \\ 4a + 2b + 2c &= -2 \end{aligned}$$

$$\Rightarrow a = 1,\ b = -2,\ c = -1$$

Thus $\mathbf{x} = \mathbf{u} - 2\mathbf{v} - \mathbf{w}$

- Notes:

A vector $\mathbf{u} = (u_1, u_2, \ldots, u_n)$ in R^n can be viewed as:

a $1 \times n$ row matrix (row vector): $\mathbf{u} = [u_1, u_2, \cdots, u_n]$

or

a $n \times 1$ column matrix (column vector): $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$

- Observation: *The matrix operations of addition and scalar multiplication give the same results as the corresponding vector operations*

Vector addition

$$\mathbf{u} + \mathbf{v} = (u_1, u_2, \cdots, u_n) + (v_1, v_2, \cdots, v_n) = (u_1 + v_1, u_2 + v_2, \cdots, u_n + v_n)$$

Scalar multiplication

$$c\mathbf{u} = c(u_1, u_2, \cdots, u_n) = (cu_1, cu_2, \cdots, cu_n)$$

Row matrix addition

$$\mathbf{u} + \mathbf{v} = [u_1, u_2, \cdots, u_n] + [v_1, v_2, \cdots, v_n] = [u_1 + v_1, u_2 + v_2, \cdots, u_n + v_n]$$

Row matrix multiplication

$$c\boldsymbol{u} = c[u_1, u_2, \cdots, u_n] = [cu_1, cu_2, \cdots, cu_n]$$

Column matrix addition

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \vdots \\ u_n + v_n \end{bmatrix}$$

Column matrix multiplication

$$c\mathbf{u} = c\begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} = \begin{bmatrix} cu_1 \\ cu_2 \\ \vdots \\ cu_n \end{bmatrix}$$

Definition

Let V be a set on which two operations (addition and scalar multiplication) are defined. If the following axioms are satisfied for every **u**, **v**, and **w** in V and every scalar (real number) c and d, then V is called a **vector space** and elements of V is called vectors

Addition:

(1) **u**+**v** is in V

(2) **u**+**v**=**v**+**u**

(3) **u**+(**v**+**w**)=(**u**+**v**)+**w**

(4) V has a an element, denoted by **0,** such that for every **u** in V, we have **u**+**0**=**u.** This element **0** is called *zero vector*.

(5) For every **u** in V, there is a vector in V denoted by –**u** such that **u**+(–**u**)=**0**

Scalar multiplication:

(6) $c\mathbf{u}$ is in V.

(7) $c(\mathbf{u}+\mathbf{v}) = c\mathbf{u} + c\mathbf{v}$

(8) $(c+d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$

(9) $c(d\mathbf{u}) = (cd)\mathbf{u}$

(10) $1(\mathbf{u}) = \mathbf{u}$

- Notes:

(1) A vector space consists of four entities:

a set of vectors, a set of scalars, and two operations

V：nonempty set

c：scalar – real number (it might be complex numbers)

$+(\mathbf{u},\mathbf{v})=\mathbf{u}+\mathbf{v}$：addition

$\bullet(c,\mathbf{u})=c\mathbf{u}$：scalar multiplication

$(V,\ +,\ \bullet)$ is called a vector space

(2) $V=\{\mathbf{0}\}$: zero vector space

- Examples of vector spaces:

(1) n-tuple space: R^n

$(u_1,u_2,\cdots,u_n)+(v_1,v_2,\cdots,v_n)=(u_1+v_1,u_2+v_2,\cdots,u_n+v_n)$ vector addition

$k(u_1,u_2,\cdots,u_n)=(ku_1,ku_2,\cdots,ku_n)$ scalar multiplication

(2) Matrix space: $V=M_{m\times n}$ (the set of all $m\times n$ matrices with real values)

Ex: $(m=n=2)$

$$\begin{bmatrix}u_{11}&u_{12}\\u_{21}&u_{22}\end{bmatrix}+\begin{bmatrix}v_{11}&v_{12}\\v_{21}&v_{22}\end{bmatrix}=\begin{bmatrix}u_{11}+v_{11}&u_{12}+v_{12}\\u_{21}+v_{21}&u_{22}+v_{22}\end{bmatrix}$$ vector addition

$$k\begin{bmatrix}u_{11}&u_{12}\\u_{21}&u_{22}\end{bmatrix}=\begin{bmatrix}ku_{11}&ku_{12}\\ku_{21}&ku_{22}\end{bmatrix}$$ scalar multiplication

(3) *n*-th degree polynomial space: $V = P_n(x)$

(the set of all real polynomials of degree $\leq n$)

Given 2 polynomials of degree n

$$p(x) = a_0 + a_1 x + \cdots + a_n x^n$$

$$q(x) = b_0 + b_1 x + \cdots + b_n x^n$$

Define:

$$p(x) + q(x) = (a_0 + b_0) + (a_1 + b_1)x + \cdots + (a_n + b_n)x^n$$

$$kp(x) = ka_0 + ka_1 x + \cdots + ka_n x^n$$

(4) Function space: $V = c(-\infty, \infty)$ (the set of all real-valued continuous functions defined on the entire real line.)

$$(f + g)(x) = f(x) + g(x)$$

$$(kf)(x) = kf(x)$$

- Thm 4.3: (Properties of scalar multiplication)

Let **v** be any element of a vector space V, and let c be any scalar. Then the following properties are true.

(1) $0\mathbf{v} = \mathbf{0}$

(2) $c\mathbf{0} = \mathbf{0}$

(3) If $c\mathbf{v} = \mathbf{0}$, then $c = 0$ or $\mathbf{v} = \mathbf{0}$

(4) $(-1)\mathbf{v} = -\mathbf{v}$

- Notes: To show that a set is not a vector space, you need only find one axiom that is not satisfied.
- Ex 6: The set V of all integer is not a vector space.

Pf:

$1 \in V, \frac{1}{2} \in R$

$(\frac{1}{2})(1) = \frac{1}{2} \notin V$ (V is not closed under scalar multiplication)

scalar, integer, noninteger

- Ex 7: The set of all second-degree polynomials is not a vector space.

Pf: Let $p(x) = x^2$ and $q(x) = -x^2 + x + 1$

$\Rightarrow p(x) + q(x) = x + 1 \notin V$

(it is not closed under vector addition)

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- Ex 8:

$V = R^2$ = the set of all ordered pairs of real numbers

We define new operations on V as follow.

vector addition: $(u_1, u_2) + (v_1, v_2) = (u_1 + v_1, u_2 + v_2)$

scalar multiplication: $c(u_1, u_2) = (cu_1, 0)$

Verify V is not a vector space.

Sol:

$\because 1(1, 1) = (1, 0) \neq (1, 1)$

$\therefore$ the set V (together with the two given operations) is not a vector space

- Subspace:

  $(V,+,\bullet)$: a vector space

  $\left.\begin{array}{l} W \neq \phi \\ W \subseteq V \end{array}\right\}$: a nonempty subset

  $(W,+,\bullet)$: a vector space (under the same operations of addition and scalar multiplication defined in V)

  $\Rightarrow$ W is a subspace of V

- Trivial subspace:

  Every vector space V has at least two subspaces.

  (1) Zero vector space $\{\mathbf{0}\}$ is a subspace of V.

  (2) V is a subspace of V.

- Thm 4.4: (Test for a subspace)

  If W is a nonempty subset of a vector space V, then W is a subspace of V if and only if the following conditions hold.

  (1) If $\mathbf{u}$ and $\mathbf{v}$ are in W, then $\mathbf{u}+\mathbf{v}$ is in W.

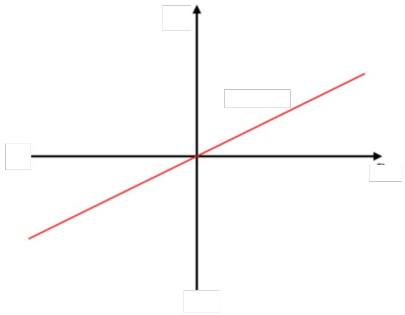
  (2) If $\mathbf{u}$ is in W and c is any scalar, then $c\mathbf{u}$ is in W.

- Ex: Subspace of R^2

  (1) $\{\mathbf{0}\}$ $\quad \mathbf{0} = (0, 0)$

  (2) Lines through the origin

  (3) R^2



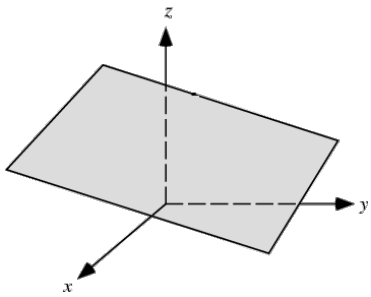
- Ex: Subspace of R^3

  (1) $\{\mathbf{0}\}$ $\quad \mathbf{0} = (0, 0, 0)$

  (2) Lines through the origin

  (3) Planes through the origin

  (4) R^3



- Ex 2: (A subspace of $M_{2\times2}$)

Let W be the set of all 2×2 symmetric matrices. Show that W is a subspace of the vector space $M_{2\times2}$, with the standard operations of matrix addition and scalar multiplication.

Sol:

$\because W \subseteq M_{2\times2}$ $\quad M_{2\times2}$: vector sapces

Let $A_1, A_2 \in W$ $\quad (A_1^T = A_1, A_2^T = A_2)$

$A_1 \in W, A_2 \in W \Rightarrow (A_1 + A_2)^T = A_1^T + A_2^T = A_1 + A_2$ $\quad (A_1 + A_2 \in W)$

$k \in R, A \in W \Rightarrow (kA)^T = kA^T = kA$ $\quad (kA \in W)$

$\therefore W$ is a subspace of $M_{2\times2}$

- Ex 3: (The set of singular matrices is not a subspace of $M_{2\times2}$)

  Let W be the set of singular matrices of order 2. Show that W is not a subspace of $M_{2\times2}$ with the standard operations.

Sol:

$$A=\begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix}\in W, B=\begin{bmatrix}0 & 0\\ 0 & 1\end{bmatrix}\in W$$

$$\therefore A+B=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}\notin W$$

$\therefore W_2$ is not a subspace of $M_{2\times2}$

- Ex 4: (The set of first-quadrant vectors is not a subspace of R^2)

  Show that $W=\{(x_1,x_2): x_1\geq 0 \text{ and } x_2\geq 0\}$, with the standard operations, is not a subspace of R^2.

Sol:

Let $\mathbf{u}=(1,1)\in W$

$\because (-1)\mathbf{u}=(-1)(1,1)=(-1,-1)\notin W$

(not closed under scalar multiplication)

$\therefore W$ is not a subspace of R^2

- Ex 6: (Determining subspaces of R^2)

Which of the following two subsets is a subspace of R^2?

(a) The set of points on the line given by $x+2y=0$.

(b) The set of points on the line given by $x+2y=1$.

Sol:

(a) $W=\{(x,y) \mid x+2y=0\}=\{(-2t,t) \mid t\in R\}$

Let $v_1=(-2t_1,t_1)\in W \quad v_2=(-2t_2,t_2)\in W$

$\because v_1+v_2=(-2(t_1+t_2),t_1+t_2)\in W$ (closed under addition)

$kv_1=(-2(kt_1),kt_1)\in W$ (closed under scalar multiplication)

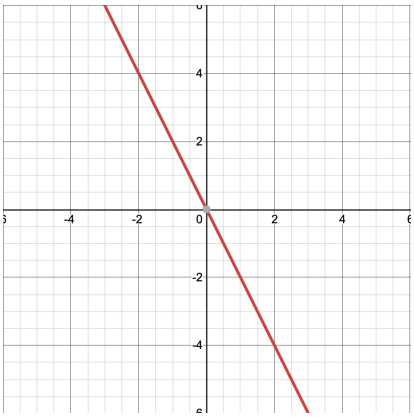
$\therefore W$ is a subspace of R^2

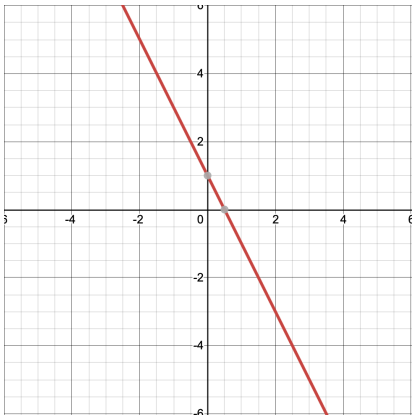
(b) $W=\{(x,y) \mid x+2y=1\}$ (Note: the zero vector is not on the line)

Let $v=(1,0)\in W$

$\because (-1)v=(-1,0)\notin W$

$\therefore W$ is not a subspace of R^2





▪ Ex 8: (Determining subspaces of R^3)

Which of the following subsets is a subspace of R^3?

(a) $W = \{(x_1, x_2, 1) \mid x_1, x_2 \in R\}$

(b) $W = \{(x_1, x_1 + x_3, x_3) \mid x_1, x_3 \in R\}$

Sol:

(a) Let $\mathbf{v} = (0,0,1) \in W$

$\Rightarrow (-1)\mathbf{v} = (0,0,-1) \notin W$

$\therefore W$ is not a subspace of R^3

(b) Let $\mathbf{v} = (v_1, v_1 + v_3, v_3) \in W$, $\mathbf{u} = (u_1, u_1 + u_3, u_3) \in W$

$\because \mathbf{v} + \mathbf{u} = (v_1 + u_1, (v_1 + u_1) + (v_3 + u_3), v_3 + u_3) \in W$

$k\mathbf{v} = (kv_1, (kv_1) + (kv_3), kv_3) \in W$

$\therefore W$ is a subspace of R^3

▪ Thm 4.5: (The intersection of two subspaces is a subspace)

If V and W are both subspaces of a vector space U, then the intersection of V and W (denoted by $V \cap W$) is also a subspace of U.

▪ Homework : Verify the above with examples from geometry of plan and space.

- Linear combination: *(recall)*

A vector $\mathbf{v}$ in a vector space V is called a linear combination of the vectors $\mathbf{u}_1, \mathbf{u}_2, \cdots, \mathbf{u}_k$ in V if $\mathbf{v}$ can be written in the form

$$\mathbf{v} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \ldots + c_k\mathbf{u}_k \qquad c_1, c_2, \cdots, c_k : \text{scalars}$$

- Ex 2-3: (Finding a linear combination)

$\mathbf{v}_1 = (1,2,3)$ $\mathbf{v}_2 = (0,1,2)$ $\mathbf{v}_3 = (-1,0,1)$

Prove (a) $\mathbf{w} = (1,1,1)$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$

(b) $\mathbf{w} = (1,-2,2)$ is not a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$

Sol:

(a) $\mathbf{w} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3$, where w $= (1,1,1)$

$$(1,1,1) = c_1(1,2,3) + c_2(0,1,2) + c_3(-1,0,1)$$
$$= (c_1 - c_3, 2c_1 + c_2, 3c_1 + 2c_2 + c_3)$$

$$\begin{aligned} c_1 \qquad - c_3 &= 1 \\ \Rightarrow\ 2c_1 + \ c_2 \qquad &= 1 \\ 3c_1 + 2c_2 + c_3 &= 1 \end{aligned}$$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 2 & 1 & 0 & 1 \\ 3 & 2 & 1 & 1 \end{array}\right] \xrightarrow{\text{Gauss–Jordan Elimination}} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$\Rightarrow c_1 = 1+t,\ c_2 = -1-2t,\ c_3 = t$

(this system has infinitely many solutions)

$t=1$

$\Rightarrow \mathbf{w} = 2\mathbf{v}_1 - 3\mathbf{v}_2 + \mathbf{v}_3$

(*b*)

$\mathbf{w} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3$, where w = (1, - 2,2)

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 2 & 1 & 0 & -2 \\ 3 & 2 & 1 & 2 \end{array}\right] \xrightarrow{\text{Gauss–Jordan Elimination}} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & -4 \\ 0 & 0 & 0 & 7 \end{array}\right]$$

$\Rightarrow$ this system has no solution $(\because 0 \neq 7)$

$\Rightarrow \mathbf{w} \neq c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3$

- the span of a set: span (S)

If $S=\{\mathbf{v}_1, \mathbf{v}_2,\ldots, \mathbf{v}_k\}$ is a set of vectors in a vector space V, then **the span of S** is the set of all linear combinations of the vectors in S,

$$span(S) = \{c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k \mid \forall c_i \in R\}$$

(the set of all linear combinations of vectors in S)

- a spanning set of a vector space:

If every vector in a given vector space V can be written as a linear combination of vectors in a given set S, then S is called **a spanning set** of the vector space V.

In other words, S is a spanning set of V if and only if

$$Span(S) = V.$$

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- Notes: In case S is a spanning set of V we use following notation

$$\text{span}(S) = V$$

$\Rightarrow$ S spans (generates) V

V is spanned (generated) by S

S is a spanning set of V

- Notes:

(1) $span(\phi) = \{\mathbf{0}\}$, we agree by consent!

(2) $S \subseteq span(S)$

(3) $S_1, S_2 \subseteq V$

$S_1 \subseteq S_2 \Rightarrow span(S_1) \subseteq span(S_2)$

- Ex 5: (A spanning set for R^3)

Show that the set $S = \{(1,2,3),(0,1,2),(-2,0,1)\}$ spans R^3

Sol:

We must determine whether an arbitrary vector $\mathbf{u} = (u_1, u_2, u_3)$ in R^3 can be written as a linear combination of $\mathbf{v}_1, \mathbf{v}_2$, and $\mathbf{v}_3$.

$$\mathbf{u} \in R^3 \Rightarrow \mathbf{u} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3$$

$$\Rightarrow \begin{aligned} c_1 \quad\quad - 2c_3 &= u_1 \\ 2c_1 + c_2 \quad\quad &= u_2 \\ 3c_1 + 2c_2 + c_3 &= u_3 \end{aligned}$$

The problem thus reduces to determining whether this system is consistent for all values of u_1, u_2, and u_3.

$$\because |A| = \begin{vmatrix} 1 & 0 & -2 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{vmatrix} \neq 0$$

$\Rightarrow A\mathbf{x} = \mathbf{b}$ has exactly one solution for every u.

$\Rightarrow span(S) = R^3$

- Thm 4.6: (Properties of Span(S))

If $S=\{\mathbf{v}_1, \mathbf{v}_2,\ldots, \mathbf{v}_k\}$ is a set of vectors in a vector space V, then

(a) span (S) is a subspace of V.

(b) Every subspace of V that contains S must contain span (S).
That is, span (S) is the smallest subspace of V that contains S.

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- Linear Independence (L.I.) and Linear Dependence (L.D.):

$S = \{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_k\}$: a set of vectors in a vector space V

We consider the equation $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k = \mathbf{0}$, where $c_1, c_2, \cdots c_k$ are unknowns.

(1) Obviously the equation has the solution ($c_1 = c_2 = \cdots = c_k = 0$)

(2) If the equation has only the trivial solution ($c_1 = c_2 = \cdots = c_k = 0$) then S is called linearly independent.

(3) If the equation has a nontrivial solution (i.e., not all zeros), then S is called linearly dependent.

- Notes:

(1) ϕ is linearly independent we agree by consent!

(2) $\mathbf{0} \in S \Rightarrow S$ is linearly dependent.

(3) $\mathbf{v} \neq \mathbf{0} \Rightarrow \{\mathbf{v}\}$ is linearly independent

(4) $S_1 \subseteq S_2$

S_1 is linearly dependent $\Rightarrow S_2$ is linearly dependent

S_2 is linearly independent $\Rightarrow S_1$ is linearly independent

- Ex 8: (Testing for linearly independent)

Determine whether the following set of vectors in R^3 is L.I. or L.D.

$$S = \{(1,2,3),(0,1,2),(-2,0,1)\}$$

Sol: $\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3$

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = \mathbf{0} \Rightarrow \begin{aligned} c_1 \quad\quad - 2c_3 &= 0 \\ 2c_1 + c_2 \quad\quad &= 0 \\ 3c_1 + 2c_2 + c_3 &= 0 \end{aligned}$$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 2 & 1 & 0 & 0 \\ 3 & 2 & 1 & 0 \end{array}\right] \xrightarrow{\text{Gauss - Jordan Elimination}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right]$$

$\Rightarrow c_1 = c_2 = c_3 = 0$ (only the trivial solution)

$\Rightarrow S$ is linearly independent

- Ex 9: (Testing for linearly independent)

Determine whether the following set of vectors in P_2 is L.I. or L.D.

$$S = \{1+x-2x^2,\ 2+5x-x^2,\ x+x^2\}$$
$$\mathbf{v}_1 \qquad \mathbf{v}_2 \qquad \mathbf{v}_3$$

Sol:

$$c_1\mathbf{v}_1+c_2\mathbf{v}_2+c_3\mathbf{v}_3 = \mathbf{0}$$

i.e. $c_1(1+x-2x^2)+c_2(2+5x-x^2)+c_3(x+x^2)=0+0x+0x^2$

$$\begin{aligned} c_1+2c_2 &= 0\\ c_1+5c_2+c_3 &= 0\\ -2c_1-c_2+c_3 &= 0 \end{aligned} \qquad \left[\begin{array}{ccc|c} 1 & 2 & 0 & 0\\ 1 & 5 & 1 & 0\\ -2 & -1 & 1 & 0 \end{array}\right] \xrightarrow{\text{G.J.}} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 0\\ 1 & 1 & \frac{1}{3} & 0\\ 0 & 0 & 0 & 0 \end{array}\right]$$

This system has infinitely many solutions.
(i.e., This system has nontrivial solutions.)

S is linearly dependent. (Ex: $c_1=2$, $c_2=-1$, $c_3=3$)

- Ex 10: (Testing for linearly independent)

Determine whether the following set of vectors in 2×2 matrix space is L.I. or L.D.

$$S = \left\{\begin{bmatrix} 2 & 1\\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 3 & 0\\ 2 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0\\ 2 & 0 \end{bmatrix}\right\}$$
$$\mathbf{v}_1 \qquad \mathbf{v}_2 \qquad \mathbf{v}_3$$

Sol:

$$c_1\mathbf{v}_1+c_2\mathbf{v}_2+c_3\mathbf{v}_3 = \mathbf{0}$$

$$c_1\begin{bmatrix} 2 & 1\\ 0 & 1 \end{bmatrix}+c_2\begin{bmatrix} 3 & 0\\ 2 & 1 \end{bmatrix}+c_3\begin{bmatrix} 1 & 0\\ 2 & 0 \end{bmatrix}=\begin{bmatrix} 0 & 0\\ 0 & 0 \end{bmatrix}$$

$$
\begin{aligned}
2c_1+3c_2+\ c_3 &= 0\\
c_1 \qquad\qquad &= 0\\
\quad 2c_2+2c_3 &= 0\\
c_1+\ c_2 \qquad &= 0
\end{aligned}
$$

$$
\left[\begin{array}{ccc|c} 2 & 3 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 2 & 2 & 0 \\ 1 & 1 & 0 & 0 \end{array}\right] \xrightarrow{\text{Gauss - Jordan Elimination}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]
$$

$c_1 = c_2 = c_3 = 0$ (This system has only the trivial solution.)

S is linearly independent.

- Thm 4.7: (A property of linearly dependent sets)

A set $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$, $k > 1$, is linearly dependent if and only if at least one of the vectors $\boldsymbol{v}_j$ in S can be written as a linear combination of the other vectors in S.

Pf:

(=>) $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \ldots + c_k\mathbf{v}_k = \mathbf{0}$

$\because$ S is linearly dependent

$\exists\ c_i \neq 0$ for some i

$$
\mathbf{v}_i = -\frac{c_1}{c_i}\mathbf{v}_1 - \cdots - \frac{c_{i-1}}{c_i}\mathbf{v}_{i-1} - \frac{c_{i+1}}{c_i}\mathbf{v}_{i+1} - \cdots - \frac{c_k}{c_i}\mathbf{v}_k
$$

($\Leftarrow$)

Let $\mathbf{v}_i = d_1\mathbf{v}_1+\ldots+d_{i-1}\mathbf{v}_{i-1}+d_{i+1}\mathbf{v}_{i+1}+\ldots+d_k\mathbf{v}_k$

$\Rightarrow$ $d_1\mathbf{v}_1+\ldots+d_{i-1}\mathbf{v}_{i-1}-\mathbf{v}_i+d_{i+1}\mathbf{v}_{i+1}+\ldots+d_k\mathbf{v}_k = \mathbf{0}$

$\Rightarrow$ $c_1=d_1, \ldots, c_{i-1}=d_{i-1}, c_i=-1, c_{i+1}=d_{i+1}, \ldots, c_k=d_k$ (nontrivial solution)

$\Rightarrow S$ is linearly dependent

- Corollary to Theorem 4.7:

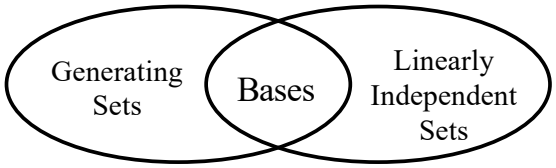
  Two vectors **u** and **v** in a vector space V are linearly dependent if and only if one is a scalar multiple of the other. That is, $\mathbf{u} = k\mathbf{v}$, for some scalar k.

4.5 Basis and Dimension

- Basis:

  V : a vector space

  $S=\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\} \subset V$



  $\begin{cases}(a)\ S \text{ spans } V \text{ (i.e., } span(S) = V) \\ (b)\ S \text{ is linearly independent}\end{cases}$

  S is called a **basis** for V

- Notes:

  (1) Ø *(empty set)* is a basis for $\{\mathbf{0}\}$ we agree by consent!

  (2) the standard basis for R^3:

  $\{i, j, k\}$ $i = (1, 0, 0),\ j = (0, 1, 0),\ k = (0, 0, 1)$

(3) the standard basis for R^n :

$\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\}$ $\mathbf{e}_1=(1,0,\ldots,0)$, $\mathbf{e}_2=(0,1,\ldots,0)$, $\mathbf{e}_n=(0,0,\ldots,1)$

Ex: R^4 $\{(1,0,0,0), (0,1,0,0), (0,0,1,0), (0,0,0,1)\}$

(4) the standard basis for $m\times n$ matrix space:

$\{ E_{ij} \mid i = 1,\ldots, m\ ,\ j = 1,\ldots .\ , n \}$, where

$\{ E_{ij} \mid 1\le i\le m\ ,\ 1\le j\le n \}$, and in E_{ij} $\begin{cases} a_{ij} = 1 \\ \text{other entries are zero} \end{cases}$

Ex: 2×2 matrix space: $\left\{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\right\}$

(5) the standard basis for $P_n(x)$:

$\{1, x, x^2, \ldots, x^n\}$

Ex: $P_3(x)$ $\{1, x, x^2, x^3\}$

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- Thm 4.8: (Uniqueness of basis representation)

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$ is a basis for a vector space V, then every vector in V can be written in one and only one way as a linear combination of vectors in S.

Pf:

$\because S$ is a basis $\Rightarrow$ $\begin{cases} 1. & span(S) = V \\ 2. & S \text{ is linearly independent} \end{cases}$

$\because span(S) = V$ Let $\mathbf{v} = c_1\mathbf{v}_1+c_2\mathbf{v}_2+\ldots+c_n\mathbf{v}_n$

$\mathbf{v} = b_1\mathbf{v}_1+b_2\mathbf{v}_2+\ldots+b_n\mathbf{v}_n$

$\Longrightarrow$ $\mathbf{0} = (c_1-b_1)\mathbf{v}_1+(c_2-b_2)\mathbf{v}_2+\ldots+(c_n-b_n)\mathbf{v}_n$

$\because S$ is linearly independent

$\Longrightarrow$ $c_1= b_1$, $c_2= b_2$,…, $c_n=$ (i.e., uniqueness)

- Thm 4.9: (Bases and linear dependence)

If $S=\{\mathbf{v}_1,\mathbf{v}_2,\cdots,\mathbf{v}_n\}$ is a basis for a vector space V, then every set containing more than n vectors in V is linearly dependent.

Pf:

Let $S_1=\{\mathbf{u}_1,\mathbf{u}_2,\ldots,\mathbf{u}_m\}$, $m>n$

$\because span(S)=V$

$$\mathbf{u}_i\in V \Rightarrow \begin{aligned}\mathbf{u}_1&=c_{11}\mathbf{v}_1+c_{21}\mathbf{v}_2+\cdots+c_{n1}\mathbf{v}_n\\ \mathbf{u}_2&=c_{12}\mathbf{v}_1+c_{22}\mathbf{v}_2+\cdots+c_{n2}\mathbf{v}_n\\ &\vdots\\ \mathbf{u}_m&=c_{1m}\mathbf{v}_1+c_{2m}\mathbf{v}_2+\cdots+c_{nm}\mathbf{v}_n\end{aligned}$$

Let $k_1\mathbf{u}_1+k_2\mathbf{u}_2+\ldots+k_m\mathbf{u}_m=\mathbf{0}$

=> $d_1\mathbf{v}_1+d_2\mathbf{v}_2+\ldots+d_n\mathbf{v}_n=\mathbf{0}$ (where $d_i=c_{i1}k_1+c_{i2}k_2+\ldots+c_{im}k_m$)

$\because S$ is L.I.

=> $d_i=0\ \forall\ i$ i.e.
$$\begin{aligned}c_{11}k_1+c_{12}k_2+\cdots+c_{1m}k_m&=0\\ c_{21}k_1+c_{22}k_2+\cdots+c_{2m}k_m&=0\\ &\vdots\\ c_{n1}k_1+c_{n2}k_2+\cdots+c_{nm}k_m&=0\end{aligned}$$

$\because$ Known: If the homogeneous system has fewer equations than variables, then it must have infinitely many solution.

$m>n$ => $k_1\mathbf{u}_1+k_2\mathbf{u}_2+\ldots+k_m\mathbf{u}_m=\mathbf{0}$ has nontrivial solution

==> S_1 is linearly dependent

- Thm 4.10: (Number of vectors in a basis)

  If a vector space V has one basis with n vectors, then every basis for V has n vectors. (All bases for a finite-dimensional vector space has the same number of vectors.)

Pf:

$S=\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$
$S'=\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m\}$ two bases for a vector space

$$\left.\begin{array}{l} S \text{ is a basis} \\ S' \text{ is L.I.} \end{array}\right\} \overset{Thm.4.9}{\Rightarrow} n \geq m$$

$$\left.\begin{array}{l} S \text{ is L.I.} \\ S' \text{ is a basis} \end{array}\right\} \overset{Thm.4.9}{\Rightarrow} n \leq m$$

$$\Rightarrow n = m$$

- Finite dimensional:

  A vector space V is called **finite dimensional**, if it has a basis consisting of a finite number of elements.

- Infinite dimensional:

  If a vector space V is not finite dimensional, then it is called **infinite dimensional**.

- Dimension:

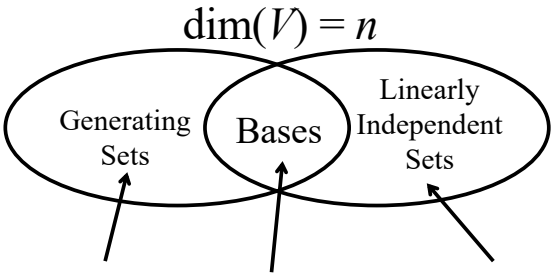
  The **dimension** of a finite dimensional vector space V is defined to be the number of vectors in a basis for V.

  V: a vector space S: a basis for V

  $\dim(V) = \#(S)$ (the number of vectors in S)

- Notes:

(1) $\dim(\{\mathbf{0}\}) = 0 = \#(\varnothing)$



(2) $\dim(V) = n$, S a subset of V $\quad \#(S) > n \quad \#(S) = n \quad \#(S) < n$

S : a generating set => $\#(S) \geq n$

S : a L.I. set => $\#(S) \leq n$

S : a basis => $\#(S) = n$

(3) $\dim(V) = n$, W is a subspace of V => $\dim(W) \leq n$

- Ex:

(1) Vector space R^n - basis $\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\}$

- $\dim(R^n) = n$

(2) Vector space $M_{m\times n}$ - basis $\{E_{ij} \mid i = 1,\ldots, m, j = 1,\ldots, n\}$

- $\dim(M_{m\times n}) = mn$

(3) Vector space $P_n(x)$ - basis $\{1, x, x^2, \ldots, x^n\}$

- $\dim(P_n(x)) = n+1$

(4) Vector space $P(x)$ - basis $\{1, x, x^2, \ldots\}$

- $\dim(P(x)) = \infty$

- Ex 9: (Finding the dimension of a subspace)

(a) $W=\{(d, c-d, c)$: c and d are real numbers$\}$

(b) $W=\{(2b, b, 0)$: b is a real number$\}$

Sol: (Note: Find a set of L.I. vectors that spans the subspace)

(a) $(d, c-d, c) = c(0, 1, 1) + d(1, -1, 0)$

$\Rightarrow S = \{(0, 1, 1), (1, -1, 0)\}$ (S is L.I. and S *spans* W)

$\Rightarrow S$ is a basis for W

$\Rightarrow \dim(W) = \#(S) = 2$

(b) $\because (2b, b, 0) = b(2, 1, 0)$

$\Rightarrow S = \{(2, 1, 0)\}$ spans W and S is L.I.

$\Rightarrow S$ is a basis for W

$\Rightarrow \dim(W) = \#(S) = 1$

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- Ex 11: (Finding the dimension of a subspace)

Let W be the subspace of all symmetric matrices in M_{2x2}.

What is the dimension of W?

Sol:

$$W = \left\{ \begin{bmatrix} a & b \\ b & c \end{bmatrix} \middle| a, b, c \in R \right\}$$

$$\because \begin{bmatrix} a & b \\ b & c \end{bmatrix} = a\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + c\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow S = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\} \text{ spans } W \text{ and } S \text{ is L.I.}$$

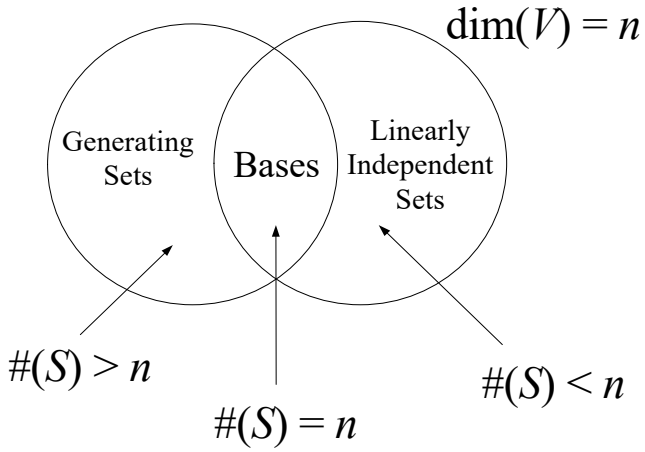
$\Rightarrow S$ is a basis for W $\Rightarrow \dim(W) = \#(S) = 3$

- Thm 4.11: (Basis tests in an n-dimensional space)

Let V be a vector space of dimension n.

(1) If $S=\{\mathbf{v}_1,\mathbf{v}_2,\cdots,\mathbf{v}_n\}$ is a linearly independent set of vectors in V, then S is a basis for V.

(2) If $S=\{\mathbf{v}_1,\mathbf{v}_2,\cdots,\mathbf{v}_n\}$ spans V, then S is a basis for V.



Recall:

- row vectors:

$$A=\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}=\begin{bmatrix} A_{(1)} \\ A_{(2)} \\ \vdots \\ A_{(m)} \end{bmatrix}$$

Row vectors of A

$$[a_{11},a_{12},\ldots,a_{1n}]=A_{(1)}$$
$$[a_{21},a_{22},\ldots,a_{2n}]=A_{(2)}$$
$$\vdots$$
$$[a_{m1},a_{m2},\ldots,a_{mn}]=A_{(m)}$$

- column vectors:

Column vectors of A

$$A=\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}=\left[A^{(1)}\vdots A^{(2)}\vdots\cdots\vdots A^{(n)}\right]$$

$$\underset{A^{(1)}}{\begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}}\quad\underset{A^{(2)}}{\begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix}}\quad\cdots\quad\underset{A^{(n)}}{\begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}}$$

Four Fundamental subspaces

Let A be an $m \times n$ matrix.

- Row space:

  The **row space** of A is the subspace of R^n spanned by the row vectors of A.

$$RS(A) = \{\alpha_1 A_{(1)} + \alpha_2 A_{(2)} + \ldots + \alpha_m A_{(m)} \mid \alpha_1, \alpha_2, \ldots, \alpha_m \in R\}$$

- Column space:

  The **column space** of A is the subspace of R^m spanned by the column vectors of A.

$$CS(A) = \{\beta_1 A^{(1)} + \beta_2 A^{(2)} + \cdots + \beta_n A^{(n)} \mid \beta_1, \beta_2, \cdots \beta_n \in R\}$$

- Null space:

  The **null space** of A is the set of all solutions of $A\mathbf{x}=\mathbf{0}$ and it is a subspace of R^n.

$$NS(A) = \{\mathbf{x} \in R^n \mid A\mathbf{x} = \mathbf{0}\}$$

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Four Fundamental subspaces

- Left Null space:

The **left null space** of A is the null space of A^T. It is the set of all solutions of $A^T\mathbf{y}=\mathbf{0}$ and is a subspace of R^m.

$$NS(A^T) = \{y \in R^m \mid A^T y = \mathbf{0}\}$$

- Compare notations

Here	In some textbook
RS(A)	$C(A^T)$
CS(A)	C(A)
NS(A)	N(A)
$NS(A^T)$	$N(A^T)$

- Recall: A $m \times n$ matrix A is row equivalent to a $m \times n$ matrix B if B can be obtained from A by applying a finite number of elementary row operations

- Thm 4.12: (Row-equivalent matrices have the same row space)

  If an $m \times n$ matrix A is row equivalent to an $m \times n$ matrix B, then the row space of A is equal to the row space of B.

- Note

  (1) The row space of a matrix is not changed by elementary row operations.

  $RS(r(A)) = RS(A)$ r: elementary row operations

  (2) Elementary row operations can change the column space.

- Thm 4.13: (Basis for the row space of a matrix)

  If a matrix A is row equivalent to a matrix B in row-echelon form, then the nonzero row vectors of B form a basis for the row space of A.

- Ex 2: (Finding a basis for a row space)

$$A=\begin{bmatrix} 1 & 3 & 1 & 3 \\ 0 & 1 & 1 & 0 \\ -3 & 0 & 6 & -1 \\ 3 & 4 & -2 & 1 \\ 2 & 0 & -4 & 2 \end{bmatrix}$$

Find a basis of row space of A

Sol:

$$A=\begin{bmatrix} 1 & 3 & 1 & 3 \\ 0 & 1 & 1 & 0 \\ -3 & 0 & 6 & -1 \\ 3 & 4 & -2 & 1 \\ 2 & 0 & -4 & 2 \end{bmatrix} \xrightarrow{Gauss\ E.} B=\begin{bmatrix} 1 & 3 & 1 & 3 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} \mathbf{w}_1 \\ \mathbf{w}_2 \\ \mathbf{w}_3 \end{matrix}$$

$$\begin{matrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 & \mathbf{a}_4 \end{matrix} \qquad \begin{matrix} \mathbf{b}_1 & \mathbf{b}_2 & \mathbf{b}_3 & \mathbf{b}_4 \end{matrix}$$

a basis for $RS(A)$ = {the nonzero row vectors of B} (Thm 4.13)
= $\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$ = $\{(1, 3, 1, 3), (0, 1, 1, 0), (0, 0, 0, 1)\}$

- Notes:

(1) $\mathbf{b}_3 = -2\mathbf{b}_1 + \mathbf{b}_2 \Rightarrow \mathbf{a}_3 = -2\mathbf{a}_1 + \mathbf{a}_2$

(2) $\{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_4\}$ is L.I. $\Rightarrow \{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_4\}$ is L.I.

- Ex 3: (Finding a basis for a subspace)

Find a basis for the subspace of R^3 spanned by

$$S = \{(-1, 2, 5), (3, 0, 3), (5, 1, 8)\} \quad (\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$$

Sol:

$$A = \begin{bmatrix} -1 & 2 & 5 \\ 3 & 0 & 3 \\ 5 & 1 & 8 \end{bmatrix} \begin{matrix} \mathbf{v}_1 \\ \mathbf{v}_2 \\ \mathbf{v}_3 \end{matrix} \xrightarrow{\text{G.E.}} B = \begin{bmatrix} 1 & -2 & -5 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix} \begin{matrix} \mathbf{w}_1 \\ \mathbf{w}_2 \end{matrix}$$

a basis for $span(\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\})$

= a basis for $RS(A)$

= {the nonzero row vectors of B} (Thm 4.13)

$= \{\mathbf{w}_1, \mathbf{w}_2\}$

$= \{(1, -2, -5), (0, 1, 3)\}$

- Ex 4: (Finding a basis for the column space of a matrix)

Find a basis for the column space of the matrix A given in Ex 2.

$$A = \begin{bmatrix} 1 & 3 & 1 & 3 \\ 0 & 1 & 1 & 0 \\ -3 & 0 & 6 & -1 \\ 3 & 4 & -2 & 1 \\ 2 & 0 & -4 & -2 \end{bmatrix}$$

Sol. 1st method:

$$A^T = \begin{bmatrix} 1 & 0 & -3 & 3 & 2 \\ 3 & 1 & 0 & 4 & 0 \\ 1 & 1 & 6 & -2 & -4 \\ 3 & 0 & -1 & 1 & -2 \end{bmatrix} \xrightarrow{G.E.} B = \begin{bmatrix} 1 & 0 & -3 & 3 & 2 \\ 0 & 1 & 9 & -5 & -6 \\ 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} \boldsymbol{w}_1 \\ \boldsymbol{w}_2 \\ \boldsymbol{w}_3 \end{matrix}$$

$\because$ $CS(A)=RS(A^T)$

$\therefore$ a basis for $CS(A)$

$=$ a basis for $RS(A^T)$

$=$ {the nonzero vectors of B}

$=\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$

$$=\left\{\begin{bmatrix}1\\0\\-3\\3\\2\end{bmatrix}, \begin{bmatrix}0\\1\\9\\-5\\-6\end{bmatrix}, \begin{bmatrix}0\\0\\1\\-1\\-1\end{bmatrix}\right\}$$ (a basis for the column space of A)

- Note: This basis is not a subset of $\{\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3, \mathbf{c}_4\}$.

- Sol. 2nd method:

$$A=\begin{bmatrix}1&3&1&3\\0&1&1&0\\-3&0&6&-1\\3&4&-2&1\\2&0&-4&-2\end{bmatrix} \xrightarrow{G.E.} B=\begin{bmatrix}1&3&1&3\\0&1&1&0\\0&0&0&1\\0&0&0&0\\0&0&0&0\end{bmatrix}$$

$\mathbf{c}_1$ $\mathbf{c}_2$ $\mathbf{c}_3$ $\mathbf{c}_4$ $\quad$ $\mathbf{v}_1$ $\mathbf{v}_2$ $\mathbf{v}_3$ $\mathbf{v}_4$

Leading 1 => $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_4\}$ is a basis for $CS(B)$

$\{\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_4\}$ is a basis for $CS(A)$

- Notes:

(1) This basis is a subset of $\{\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3, \mathbf{c}_4\}$.

(2) $\mathbf{v}_3 = -2\mathbf{v}_1 + \mathbf{v}_2$, thus $\mathbf{c}_3 = -2\mathbf{c}_1 + \mathbf{c}_2$.

- Thm 4.14: (Solutions of a homogeneous system)

If A is an $m \times n$ matrix, then the set of all solutions of the homogeneous system of linear equations $A\mathbf{x} = \mathbf{0}$ is a subspace of R^n called the nullspace of A. $NS(A) = \{x \in R^n \mid Ax = 0\}$

Pf:

$NS(A) \subseteq R^n$

$NS(A) \neq \phi \quad (\because A\mathbf{0} = \mathbf{0})$

Let $\mathbf{x}_1, \mathbf{x}_2 \in NS(A) \quad (\text{i.e. } A\mathbf{x}_1 = \mathbf{0}, A\mathbf{x}_2 = \mathbf{0})$

Then $(1)A(\mathbf{x}_1 + \mathbf{x}_2) = A\mathbf{x}_1 + A\mathbf{x}_2 = \mathbf{0} + \mathbf{0} = \mathbf{0}$ Addition

$(2)A(c\mathbf{x}_1) = c(A\mathbf{x}_1) = c(\mathbf{0}) = \mathbf{0}$ Scalar multiplication

Thus $NS(A)$ is a subspace of R^n

- Notes: The nullspace of A is also called the solution space of the homogeneous system $A\mathbf{x} = \mathbf{0}$.

- Ex 6: (Finding the solution space of a homogeneous system)

Find the nullspace of the matrix A.

$$A = \begin{bmatrix} 1 & 2 & -2 & 1 \\ 3 & 6 & -5 & 4 \\ 1 & 2 & 0 & 3 \end{bmatrix}$$

Sol: The nullspace of A is the solution space of $A\mathbf{x} = \mathbf{0}$.

$$A = \begin{bmatrix} 1 & 2 & -2 & 1 \\ 3 & 6 & -5 & 4 \\ 1 & 2 & 0 & 3 \end{bmatrix} \xrightarrow{G.J.E} \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

x_2 and x_4 are free variables

$$\Rightarrow x_1 = -2s - 3t, x_2 = s, x_3 = -t, x_4 = t$$

$$\Rightarrow \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -2s - 3t \\ s \\ -t \\ t \end{bmatrix} = s\begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t\begin{bmatrix} -3 \\ 0 \\ -1 \\ 1 \end{bmatrix} = s\mathbf{v}_1 + t\mathbf{v}_2$$

$\rightarrow NS(A) = \{s\mathbf{v}_1 + t\mathbf{v}_2 \mid s, t \in R\}$

- Thm 4.15: (Row and column space have equal dimensions)

  If A is an $m \times n$ matrix, then the row space and the column space of A have the same dimension.

$$\dim(RS(A)) = \dim(CS(A))$$

- Rank:

  The dimension of the row (or column) space of a matrix A is called the **rank** of A and is denoted by $\text{rank}(A)$.

$$\text{rank}(A) = \dim(RS(A)) = \dim(CS(A))$$

- Nullity:

  The dimension of the nullspace of A is called the nullity of A.

$$\text{nullity}(A) = \dim(NS(A))$$

- Note: $\text{rank}(A^T) = \text{rank}(A)$

  Pf: $\text{rank}(A^T) = \dim(RS(A^T)) = \dim(CS(A)) = \text{rank}(A)$

- Thm 4.16: (Dimension of the solution space)

  If A is an $m \times n$ matrix of rank r, then the dimension of the solution space of $A\mathbf{x} = \mathbf{0}$ is $n - r$. That is

$$n = \text{rank}(A) + \text{nullity}(A)$$

- Notes:

(1) rank(A): The number of leading variables in the solution of $A\mathbf{x}=\mathbf{0}$.

(The number of nonzero rows in the row-echelon form of A)

(2) nullity (A): The number of free variables in the solution of $A\mathbf{x} = \mathbf{0}$.

- Notes:

If A is an $m \times n$ matrix and rank(A) = r, then

Fundamental Subspaces	Dimension
$RS(A)=CS(A^T)$	r
$CS(A)=RS(A^T)$	r
$NS(A)$	$n - r$
$NS(A^T)$	$m - r$

- Ex 7: (Rank and nullity of a matrix)

Let the column vectors of the matrix A be denoted by $\mathbf{a}_1$, $\mathbf{a}_2$, $\mathbf{a}_3$, $\mathbf{a}_4$, and $\mathbf{a}_5$.

$$A=\begin{bmatrix} 1 & 0 & -2 & 1 & 0 \\ 0 & -1 & -3 & 1 & 3 \\ -2 & -1 & 1 & -1 & 3 \\ 0 & 3 & 9 & 0 & -12 \end{bmatrix}$$
$$\mathbf{a}_1 \quad \mathbf{a}_2 \quad \mathbf{a}_3 \quad \mathbf{a}_4 \quad \mathbf{a}_5$$

(a) Find the rank and nullity of A.

(b) Find a subset of the column vectors of A that forms a basis for the column space of A.

(c) If possible, write the third column of A as a linear combination of the first two columns.

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Sol: Let B be the reduced row-echelon form of A.

$$A=\begin{bmatrix} 1 & 0 & -2 & 1 & 0 \\ 0 & -1 & -3 & 1 & 3 \\ -2 & -1 & 1 & -1 & 3 \\ 0 & 3 & 9 & 0 & -12 \end{bmatrix} \qquad B=\begin{bmatrix} 1 & 0 & -2 & 0 & 1 \\ 0 & 1 & 3 & 0 & -4 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$
$$\mathbf{a}_1 \quad \mathbf{a}_2 \quad \mathbf{a}_3 \quad \mathbf{a}_4 \quad \mathbf{a}_5 \qquad \mathbf{b}_1 \quad \mathbf{b}_2 \quad \mathbf{b}_3 \quad \mathbf{b}_4 \quad \mathbf{b}_5$$

(a) $\text{rank}(A) = 3$ (the number of nonzero rows in B)

$\text{nullity}(A) = n - \text{rank}(A) = 5\text{-}3 = 2$

(b) Leading 1

$\Rightarrow \{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_4\}$ is a basis for $CS(B)$

$\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_4\}$ is a basis for $CS(A)$

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ 0 \\ -2 \\ 0 \end{bmatrix}, \ \mathbf{a}_2 = \begin{bmatrix} 0 \\ -1 \\ -1 \\ 3 \end{bmatrix}, \text{ and } \mathbf{a}_4 = \begin{bmatrix} 1 \\ 1 \\ -1 \\ 0 \end{bmatrix},$$

(c) $\mathbf{b}_3 = -2\mathbf{b}_1 + 3\mathbf{b}_2 \Rightarrow \quad \mathbf{a}_3 = -2\mathbf{a}_1 + 3\mathbf{a}_2$

- Thm 4.17: (Solutions of a nonhomogeneous linear system)

If $\mathbf{x}_p$ is a particular solution of the nonhomogeneous system $A\mathbf{x} = \boldsymbol{b}$, then every solution of this system can be written in the form $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h$, where $\mathbf{x}_h$ is a solution of the corresponding homogeneous system $A\mathbf{x} = \mathbf{0}$.

Pf: Let $\mathbf{x}$ be any solution of $A\mathbf{x} = \boldsymbol{b}$.

$\Rightarrow A(\mathbf{x} - \mathbf{x}_p) = A\mathbf{x} - A\mathbf{x}_p = \mathbf{b} - \mathbf{b} = \mathbf{0}.$

$\Rightarrow (\mathbf{x} - \mathbf{x}_p)$ is a solution of $A\mathbf{x} = \mathbf{0}$

Let $\boldsymbol{x}_h = \boldsymbol{x} - \boldsymbol{x}_p$

$\Rightarrow \boldsymbol{x} = \boldsymbol{x}_p + \boldsymbol{x}_h$

- Ex 8: (Finding the solution set of a nonhomogeneous system)

  Find the set of all solution vectors of the system of linear equations.

$$\begin{aligned} x_1 \quad\quad\quad - 2x_3 + x_4 &= 5 \\ 3x_1 + x_2 - 5x_3 \quad\quad &= 8 \\ x_1 + 2x_2 \quad\quad - 5x_4 &= -9 \end{aligned}$$

Sol:

$$\left[\begin{array}{cccc:c} 1 & 0 & -2 & 1 & 5 \\ 3 & 1 & -5 & 0 & 8 \\ 1 & 2 & 0 & -5 & -9 \end{array}\right] \xrightarrow{G.J.E} \left[\begin{array}{cccc:c} 1 & 0 & -2 & 1 & 5 \\ 0 & 1 & 1 & -3 & -7 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

s t

$$\Rightarrow \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2s - t + 5 \\ -s + 3t - 7 \\ s + 0t + 0 \\ 0s + t + 0 \end{bmatrix} = s\begin{bmatrix} 2 \\ -1 \\ 1 \\ 0 \end{bmatrix} + t\begin{bmatrix} -1 \\ 3 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 5 \\ -7 \\ 0 \\ 0 \end{bmatrix}$$

$$= s\mathbf{u}_1 + t\mathbf{u}_2 + \mathbf{x}_p$$

i.e. $\mathbf{x}_p = \begin{bmatrix} 5 \\ -7 \\ 0 \\ 0 \end{bmatrix}$ is a particular solution vector of $A\mathbf{x}=\mathbf{b}$.

$\mathbf{x}_h = s\mathbf{u}_1 + t\mathbf{u}_2$ is a solution of $A\mathbf{x} = \mathbf{0}$

- Thm 4.18: (Solution of a system of linear equations)

The system of linear equations $A\mathbf{x} = \mathbf{b}$ is consistent if and only if $\mathbf{b}$ is in the column space of A.

Pf.

Let

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad \boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \text{and} \quad \boldsymbol{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

be the coefficient matrix, the column matrix of unknowns, and the right-hand side, respectively, of the system $A\mathbf{x} = \mathbf{b}$.

Then

$$A\mathbf{x} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \end{bmatrix}$$

$$= x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}.$$

So, $\mathbf{b} = A\mathbf{x} = x_1A^{(1)} + x_2A^{(2)} + \ldots + x_nA^{(n)}$, which is a linear combination of column vector of A.

Hence, $A\mathbf{x} = \mathbf{b}$ is consistent if and only if $\mathbf{b}$ is a linear combination of the columns of A. That is, the system is consistent if and only if $\mathbf{b}$ is in the subspace of R^m spanned by the columns of A.

- Note:

  If rank([A|**b**])=rank(A)

  Then the system $A\mathbf{x}=\mathbf{b}$ is consistent.

- Ex 9: (Consistency of a system of linear equations)

$$\begin{aligned} x_1 + x_2 - x_3 &= -1 \\ x_1 + x_3 &= 3 \\ 3x_1 + 2x_2 - x_3 &= 1 \end{aligned}$$

Sol:

$$A=\begin{bmatrix} 1 & 1 & -1 \\ 1 & 0 & 1 \\ 3 & 2 & -1 \end{bmatrix} \xrightarrow{G.J.E.} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$[A\vdots\mathbf{b}]=\left[\begin{array}{ccc:c} 1 & 1 & -1 & -1 \\ 1 & 0 & 1 & 3 \\ 3 & 2 & -1 & 1 \end{array}\right] \xrightarrow{G.J.E.} \left[\begin{array}{ccc:c} 1 & 0 & 1 & 3 \\ 0 & 1 & -2 & -4 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$\mathbf{c}_1\ \mathbf{c}_2\ \mathbf{c}_3\ \mathbf{b}$ $\qquad$ $\mathbf{w}_1\ \mathbf{w}_2\ \mathbf{w}_3\ \mathbf{v}$

$\because \mathbf{v}=3\mathbf{w}_1-4\mathbf{w}_2$

$\Rightarrow \mathbf{b}=3\mathbf{c}_1-4\mathbf{c}_2+0\mathbf{c}_3$ ($\mathbf{b}$ is in the column space of A)

$\Rightarrow$ The system of linear equations is consistent.

- Check:

$$\text{rank}(A)=\text{rank}([A\ \vdots\ \mathbf{b}])=2$$

- Summary of equivalent conditions for square matrices:

If A is an $n \times n$ matrix, then the following conditions are equivalent.

(1) A is invertible

(2) $A\mathbf{x} = \mathbf{b}$ has a unique solution for any $n \times 1$ matrix $\mathbf{b}$.

(3) $A\mathbf{x} = \mathbf{0}$ has only the trivial solution

(4) $\text{rank}(A) = n$

(5) The n row vectors of A are linearly independent.

(6) The n column vectors of A are linearly independent.

4.7 Coordinates and Change of Basis

- Coordinate representation relative to a basis

Let $B = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ be an ordered basis for a vector space V and let $\mathbf{x}$ be a vector in V such that

$$\mathbf{x} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n.$$

The scalars $c_1, c_2, \ldots, c_n$ are called the **coordinates of x relative to the basis B.** The **coordinate matrix** (or **coordinate vector**) of $\mathbf{x}$ relative to B is the column matrix in R^n whose components are the coordinates of $\mathbf{x}$.

$$[\mathbf{x}]_B = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

- Ex 1: (Coordinates and components in R^n)

Find the coordinate matrix of $\mathbf{x} = (-2, 1, 3)$ in R^3

relative to the standard basis

$$S = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$$

Sol:

$$\because \mathbf{x} = (-2, 1, 3) = -2(1, 0, 0) + 1(0, 1, 0) + 3(0, 0, 1),$$

$$\therefore [\mathbf{x}]_S = \begin{bmatrix} -2 \\ 1 \\ 3 \end{bmatrix}.$$

- Ex 3: (Finding a coordinate matrix relative to a nonstandard basis)

Find the coordinate matrix of $\mathbf{x}=(1, 2, -1)$ in R^3

relative to the (nonstandard) basis

$$B' = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} = \{(1, 0, 1), (0, -1, 2), (2, 3, -5)\}$$

Sol: $\boldsymbol{x} = c_1\boldsymbol{u}_1 + c_2\boldsymbol{u}_2 + c_3\boldsymbol{u}_3 \Rightarrow (1, 2, -1) = c_1(1, 0, 1) + c_2(0, -1, 2) + c_3(2, 3, -5)$

$$\Rightarrow \begin{array}{rcrcrcl} c_1 & & & + & 2c_3 & = & 1 \\ & & -c_2 & + & 3c_3 & = & 2 \\ c_1 & + & 2c_2 & - & 5c_3 & = & -1 \end{array} \quad \text{i.e.} \quad \begin{bmatrix} 1 & 0 & 2 \\ 0 & -1 & 3 \\ 1 & 2 & -5 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & -1 & 3 & 2 \\ 1 & 2 & -5 & -1 \end{bmatrix} \xrightarrow{\text{G.J.E.}} \begin{bmatrix} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & -8 \\ 0 & 0 & 1 & -2 \end{bmatrix} \Rightarrow [\boldsymbol{x}]_{B'} = \begin{bmatrix} 5 \\ -8 \\ -2 \end{bmatrix}$$

- Change of basis problem:

  You were given the coordinates of a vector relative to one basis B;

  and you are asked to find the coordinates of the same vector relative to another basis B'.

- Ex: (Change of basis)

  Consider two bases for a vector space V

$$B = \{\mathbf{u}_1, \mathbf{u}_2\},\ B' = \{\mathbf{u}'_1, \mathbf{u}'_2\}$$

$$\text{If } [\mathbf{u}'_1]_B = \begin{bmatrix} a \\ b \end{bmatrix},\ [\mathbf{u}'_2]_B = \begin{bmatrix} c \\ d \end{bmatrix}$$

$$\text{i.e., } \mathbf{u}'_1 = a\mathbf{u}_1 + b\mathbf{u}_2,\quad \mathbf{u}'_2 = c\mathbf{u}_1 + d\mathbf{u}_2$$

$$\text{Let } \mathbf{v} \in V,\ [\mathbf{v}]_{B'} = \begin{bmatrix} k_1 \\ k_2 \end{bmatrix}$$

$$\begin{aligned} \Rightarrow \mathbf{v} &= k_1\mathbf{u}'_1 + k_2\mathbf{u}'_2 \\ &= k_1(a\mathbf{u}_1 + b\mathbf{u}_2) + k_2(c\mathbf{u}_1 + d\mathbf{u}_2) \\ &= (k_1a + k_2c)\mathbf{u}_1 + (k_1b + k_2d)\mathbf{u}_2 \end{aligned}$$

So, coordinates of $\mathbf{v}$ relative to the basis $B=\{\mathbf{u}_1,\mathbf{u}_2\}$ are k_1a+k_2c, k_1b+k_2d. That is,

$$\begin{aligned} \Rightarrow [\mathbf{v}]_B &= \begin{bmatrix} k_1a + k_2c \\ k_1b + k_2d \end{bmatrix} = \begin{bmatrix} a & c \\ b & d \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \end{bmatrix} \\ &= \left[[\mathbf{u}'_1]_B\ [\mathbf{u}'_2]_B \right] [\mathbf{v}]_{B'} \end{aligned}$$

- Transition matrix from B' to B:

Let $B=\{\mathbf{u}_1,\mathbf{u}_2,...,\mathbf{u}_n\}$ and $B'=\{\mathbf{u}'_1,\mathbf{u}'_2...,\mathbf{u}'_n\}$ be two bases for a vector space V

If $[\mathbf{v}]_B$ is the coordinate matrix of $\mathbf{v}$ relative to B

$[\mathbf{v}]_{B'}$ is the coordinate matrix of $\mathbf{v}$ relative to B'

then $[\mathbf{v}]_B = P[\mathbf{v}]_{B'}$

$$= \left[[\mathbf{u}'_1]_B, [\mathbf{u}'_2]_B, ..., [\mathbf{u}'_n]_B\right][v]_{B'}$$

where

$$P = \left[[\mathbf{u}'_1]_B, [\mathbf{u}'_2]_B, ..., [\mathbf{u}'_n]_B\right]$$

is called **the transition matrix from B' to B**

- Remark: The bases B and B' have similar role. Therefore, we can exchange B and B' and we have
- Thm 4.19: (The inverse of a transition matrix)

If P is the transition matrix from a basis B' to a basis B in R^n, then

(1) P is invertible

(2) The transition matrix from B to B' is P^{-1}

- Notes:

$$B=\{\mathbf{u}_1,\mathbf{u}_2,...,\mathbf{u}_n\},\quad B'=\{\mathbf{u}'_1,\mathbf{u}'_2,...,\mathbf{u}'_n\}$$

$$[\mathbf{v}]_B = \left[[\mathbf{u}'_1]_B, [\mathbf{u}'_2]_B, ..., [\mathbf{u}'_n]_B\right][\mathbf{v}]_{B'} = P[\mathbf{v}]_{B'}$$

$$[\mathbf{v}]_{B'} = \left[[\mathbf{u}_1]_{B'}, [\mathbf{u}_2]_{B'}, ..., [\mathbf{u}_n]_{B'}\right][\mathbf{v}]_B = P^{-1}[\mathbf{v}]_B$$

- Thm 4.20: (Transition matrix from B to B')

  Let $B=\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ and $B' =\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ be two bases for R^n. Then the transition matrix P from B' to B can be found by using Gauss-Jordan elimination on the $n\times 2n$ matrix $[B \vdots \mathrm{B}']$ as follows.

$$[B \vdots \mathrm{B}'] \longrightarrow [\mathrm{I}_n \vdots \mathrm{P}]$$

- Ex 5: (Finding a transition matrix)

  $B=\{(-3, 2), (4,-2)\}$ and $B' =\{(-1, 2), (2,-2)\}$ are two bases for R^2

  (a) Find the transition matrix from B' to B.

  (b) Let $[\mathbf{v}]_{B'} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, find $[\mathbf{v}]_B$

  (c) Find the transition matrix from B to B' .

Sol:

(a) $\left[\begin{array}{cc:cc} -3 & 4 & -1 & 2 \\ 2 & -2 & 2 & -2 \end{array}\right] \xrightarrow{\text{G.J.E.}} \left[\begin{array}{cc:cc} 1 & 0 & 3 & -2 \\ 0 & 1 & 2 & -1 \end{array}\right]$

$B \qquad B' \qquad\qquad I \qquad P$

$\therefore P = \begin{bmatrix} 3 & -2 \\ 2 & -1 \end{bmatrix}$ (the transition matrix from B' to B)

(b)

Check : $[\boldsymbol{v}]_{B'} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \Rightarrow [\boldsymbol{v}]_B = P[\boldsymbol{v}]_{B'} = \begin{bmatrix} 3 & -2 \\ 2 & -1 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$

$[\boldsymbol{v}]_{B'} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \Rightarrow \boldsymbol{v} = (1)(-1,2) + (2)(2,-2) = (3,-2)$

$[\boldsymbol{v}]_B = \begin{bmatrix} -1 \\ 0 \end{bmatrix} \Rightarrow \boldsymbol{v} = (-1)(3,-2) + (0)(4,-2) = (3,-2)$

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(c)

$\left[\begin{array}{cc:cc} -1 & 2 & -3 & 4 \\ 2 & -2 & 2 & -2 \end{array}\right] \xrightarrow{\text{G.J.E.}} \left[\begin{array}{cc:cc} 1 & 0 & -1 & 2 \\ 0 & 1 & -2 & 3 \end{array}\right]$

$B' \qquad B \qquad\qquad I \qquad P^{-1}$

$\therefore P^{-1} = \begin{bmatrix} -1 & 2 \\ -2 & 3 \end{bmatrix}$ (the transition matrix from B *to* B')

- Check:

$$PP^{-1} = \begin{bmatrix} 3 & -2 \\ 2 & -1 \end{bmatrix}\begin{bmatrix} -1 & 2 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$$

- Ex 6: (Coordinate representation in $P_3(x)$)

(a) Find the coordinate matrix of $p = 3x^3-2x^2+4$ relative to the standard basis $S = \{1, x, x^2, x^3\}$ in $P_3(x)$.

(b) Find the coordinate matrix of $p = 3x^3-2x^2+4$ relative to the basis $S = \{1, 1+x, 1+x^2, 1+x^3\}$ in $P_3(x)$.

Sol:

(a) $p = (4)(1)+(0)(x)+(-2)(x^2)+(3)(x^3) \Rightarrow [p]_B = \begin{bmatrix} 4 \\ 0 \\ -2 \\ 3 \end{bmatrix}$

(b) $p = (3)(1)+(0)(1+x)+(-2)(1+x^2)+(3)(1+x^3) \Rightarrow [p]_B = \begin{bmatrix} 3 \\ 0 \\ -2 \\ 3 \end{bmatrix}$

- Ex: (Coordinate representation in M_{2x2})

Find the coordinate matrix of $x = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$ relative to the standard basis in M_{2x2}.

$$B = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

Sol:

$$x = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = 5\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + 6\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + 7\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + 8\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow [x]_B = \begin{bmatrix} 5 \\ 6 \\ 7 \\ 8 \end{bmatrix}$$